***Solution:***

| *Class* | *f* | *cf* |
|---|---|---|
| 0 – 5 | 18 | 18 |
| 5 – 10 | 20 | 38 |
| 10 – 15 | 36 | 74 |
| 15 – 20 | 40 | 114 |
| 20 – 25 | 26 | 140 |
| 25 – 30 | 16 | 156 |
| | N=156 | |

$$\frac{61N}{100} = \left(\frac{61 \times 156}{100}\right)$$

$$= 95.16 \text{ item}$$

This item is the interval 15-20. Thus

$l = 15, m = 74, f = 40, c = 5$

$$P_{61} = l + \left(\frac{\frac{61N}{100} - m}{f}\right) \times c$$

$$= 15 + \left(\frac{95.16 - 74}{40}\right) \times 5$$

$$= 15 + \frac{21.16}{40} \times 5$$

$$= 17.645$$

$P_{61} = 17.645$

**Points to Remember**

- A central tendency is a single figure that represents the whole mass of data
- Arithmetic mean or mean is the number which is obtained by adding the values of all the items of a series and dividing the total by the number of items.
- When all items of a series are given equal importance than it is called simple arithmetical mean and when different items of a series are given different weights according with their relative importance is known as weighted arithmetic mean.



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